

CSE421

Computer Networks Project

Title: Bangladesh Board Examination Emergency Network (BBEEN)

Group ID: 2302

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Assumptions

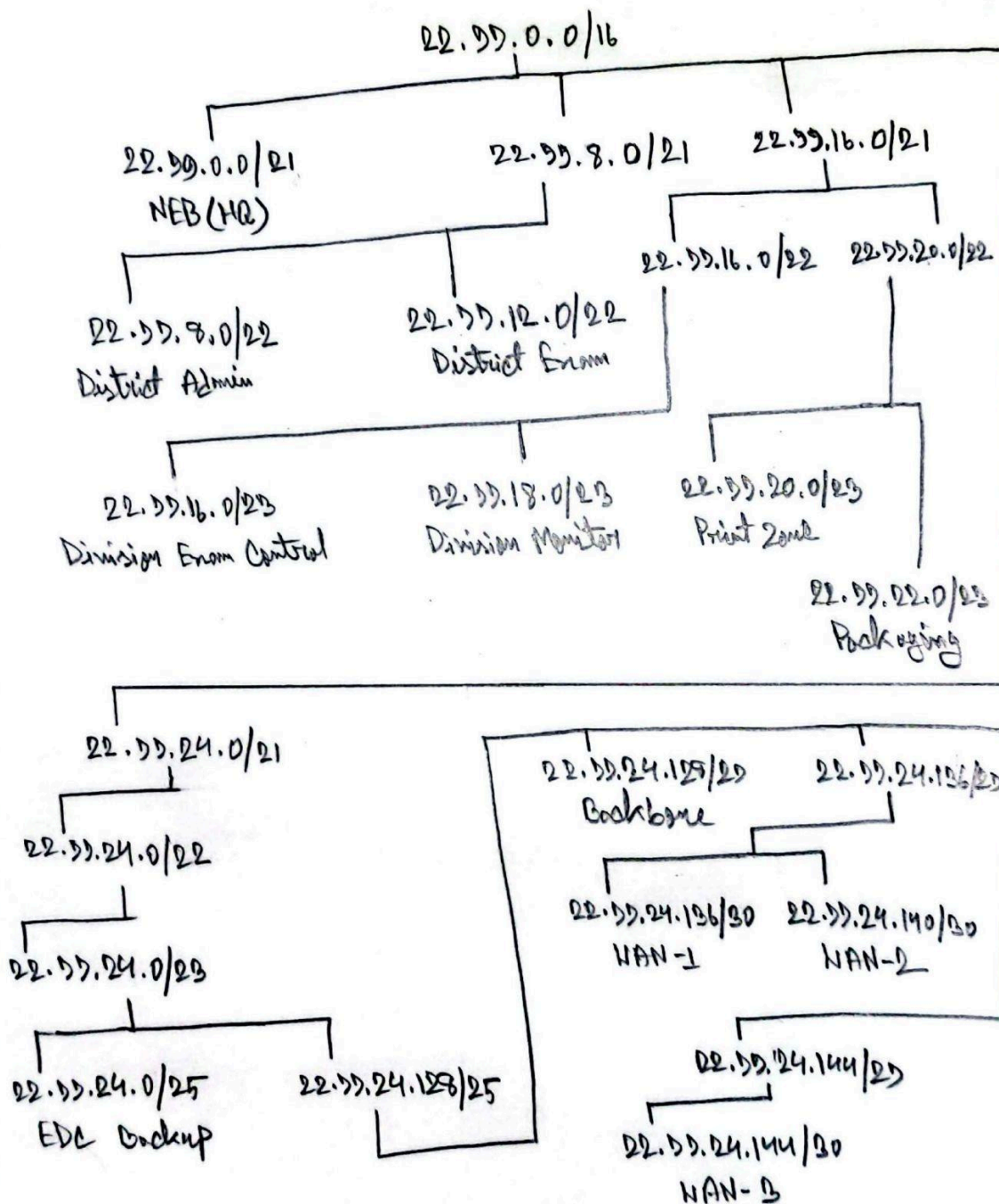
The following assumptions were made during the design and implementation of the BBEEN network:

1. **Host Distribution:** The project table provides total device counts per main unit but does not specify distribution between sub-units. We assumed an equal 50/50 split:
 - Divisional Board Office: 500 devices (Exam Control) + 500 devices (Monitoring)
 - District Exam Center: 600 devices (Admin) + 600 devices (Exam Cell)
 - Secure Printing Press: 350 devices (Print Zone) + 350 devices (Packaging)
2. **Server Placement:** The NEB-Mail-Server (mail.neb.gov.bd) is implemented as a separate server at 22.99.0.3 on the National Education Board (HQ) LAN. Same for the Print-Mail-Server (mail.print.gov.bd), which is implemented as a separate server at 22.99.20.3 on the Secure Printing Press LAN.
3. **Packet Tracer Server Limitation:** Generic servers in Packet Tracer do not support Cisco IOS CLI. DNS, HTTP, DHCP, and Email services were configured via the GUI Services tab.
4. **Backbone Switch Port Usage:** The 2960-24TT switch has only 2 GigabitEthernet ports. DIV-Router and DIS-Router connect via Gig0/1 and Gig0/2; PRINT-Router connects via Fa0/1.
5. **WAN Clocking:** NEB-Router acts as the DCE side for all serial WAN links with a clock rate of 64000.
6. **Excluded Address:** Do not mention the actual number of reserved hosts needed in the network. For this reason, the first ten IP addresses were excluded during the DHCP setup.

VLSM table:

Network	Host	Host + 2	2ⁿ Form	Host Bit	Network Bit
NEB (HQ)	2000	2002	2048	11	21
District Admin	600	602	1024	10	22
District Exam	600	602	1024	10	22
Division Exam Control	500	502	512	9	23
Division Monitor Cell	500	502	512	9	23
Print Zone	350	352	512	9	23
Packaging	350	352	512	9	23
EDC	120	122	128	7	25
Backbone Switch	3	5	8	3	29
WAN-1	2	4	4	2	30
WAN-2	2	4	4	2	30
WAN-3	2	4	4	2	30

Base Address - 22.55.0.0/16



IP Addressing Table:

National Education Board (HQ) - 22.99.0.0/21 (Static IPs)

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
NEB-Router	GigabitEthernet0/0	22.99.0.1	255.255.248.0	N/A
NEB-Router	Serial0/0/0	22.99.24.137	255.255.255.252	N/A
NEB-Router	Serial0/0/1	22.99.24.141	255.255.255.252	N/A
NEB-Router	Serial0/1/0	22.99.24.145	255.255.255.252	N/A
NEB-DNS-Web-Server	FastEthernet0	22.99.0.2	255.255.248.0	22.99.0.1
NEB-Mail-Server	FastEthernet0	22.99.0.3	255.255.248.0	22.99.0.1
NEB-PC1	FastEthernet0	22.99.0.10	255.255.248.0	22.99.0.1
NEB-PC2	FastEthernet0	22.99.0.11	255.255.248.0	22.99.0.1
NEB-Printer	FastEthernet0	22.99.0.12	255.255.248.0	22.99.0.1

District Exam Centre - 22.99.8.0/22 and 22.99.12.0/22 (Router DHCP)

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
DIS-Router	GigabitEthernet 0/0	22.99.8.1	255.255.252.0	N/A
DIS-Router	GigabitEthernet 0/1	22.99.12.1	255.255.252.0	N/A
DIS-Router	GigabitEthernet 0/2	22.99.24.130	255.255.255.248	N/A
DIS-Admin-PC1	FastEthernet0	DHCP	255.255.252.0	22.99.8.1
DIS-Admin-PC2	FastEthernet0	DHCP	255.255.252.0	22.99.8.1
DIS-Admin-Printer	FastEthernet0	DHCP	255.255.252.0	22.99.8.1

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
DIS-Exam-PC1	FastEthernet0	DHCP	255.255.252.0	22.99.12.1
DIS-Exam-PC2	FastEthernet0	DHCP	255.255.252.0	22.99.12.1
DIS-Exam-Printer	FastEthernet0	DHCP	255.255.252.0	22.99.12.1

Divisional Board Office - 22.99.16.0/23 and 22.99.18.0/23 (Router DHCP)

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
DIV-Router	GigabitEthernet0 /0	22.99.16.1	255.255.254.0	N/A
DIV-Router	GigabitEthernet0 /1	22.99.18.1	255.255.254.0	N/A
DIV-Router	GigabitEthernet0 /2	22.99.24.129	255.255.255.248	N/A
DIV-Router	Serial0/0/0	22.99.24.138	255.255.255.252	N/A
DIV-Router	Serial0/0/1	22.99.24.142	255.255.255.252	N/A
DIV-Exam-PC1	FastEthernet0	DHCP	255.255.254.0	22.99.16.1
DIV-Exam-PC2	FastEthernet0	DHCP	255.255.254.0	22.99.16.1
DIV-Exam-Printer	FastEthernet0	DHCP	255.255.254.0	22.99.16.1
DIV-Mon-PC1	FastEthernet0	DHCP	255.255.254.0	22.99.18.1
DIV-Mon-PC2	FastEthernet0	DHCP	255.255.254.0	22.99.18.1
DIV-Mon-Printer	FastEthernet0	DHCP	255.255.254.0	22.99.18.1

Secure Printing Press - 22.99.20.0/23 and 22.99.22.0/23 (Server DHCP)

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
PRINT-Router	GigabitEthernet0/0	22.99.20.1	255.255.254.0	N/A

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
PRINT-Router	GigabitEthernet0/1	22.99.22.1	255.255.254.0	N/A
PRINT-Router	GigabitEthernet0/2	22.99.24.131	255.255.255.248	N/A
PRINT-DHCP-Server	FastEthernet0	22.99.20.2	255.255.254.0	22.99.20.1
PRINT-Mail-Server	FastEthernet0	22.99.20.3	255.255.254.0	22.99.20.1
PRINT-Zone-PC1	FastEthernet0	DHCP	255.255.254.0	22.99.20.1
PRINT-Zone-PC2	FastEthernet0	DHCP	255.255.254.0	22.99.20.1
PRINT-Zone-Printer	FastEthernet0	DHCP	255.255.254.0	22.99.20.1
PRINT-Pack-PC1	FastEthernet0	DHCP	255.255.254.0	22.99.22.1
PRINT-Pack-PC2	FastEthernet0	DHCP	255.255.254.0	22.99.22.1
PRINT-Pack-Printer	FastEthernet0	DHCP	255.255.254.0	22.99.22.1

Emergency Distribution Centre - 22.99.24.0/25 (Static only)

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
EDC-Router	GigabitEthernet0/0	22.99.24.1	255.255.255.128	N/A
EDC-Router	Serial0/0/0	22.99.24.146	255.255.255.252	N/A
EDC-PC1	FastEthernet0	22.99.24.10	255.255.255.128	22.99.24.1
EDC-PC2	FastEthernet0	22.99.24.11	255.255.255.128	22.99.24.1
EDC-Printer	FastEthernet0	22.99.24.12	255.255.255.128	22.99.24.1

Router and Server Configuration Commands and Steps

Base Network: 22.99.0.0/16

Router Part:

NEB-Router

enable

configure terminal

hostname NEB-Router

#LAN: NEB HQ (22.99.0.0/21)

interface GigabitEthernet0/0

ip address 22.99.0.1 255.255.248.0

no shutdown

#WAN Primary to DIV-Router (22.99.24.136/30)

interface Serial0/0/0

ip address 22.99.24.137 255.255.255.252

no shutdown

clock rate 64000

#WAN Backup to DIV-Router (22.99.24.140/30)

interface Serial0/0/1

ip address 22.99.24.141 255.255.255.252

no shutdown

clock rate 64000

#WAN to EDC-Router (22.99.24.144/30)

interface Serial0/1/0

ip address 22.99.24.145 255.255.255.252

no shutdown

clock rate 64000

#STATIC ROUTES

#Next-hop static to EDC

ip route 22.99.24.0 255.255.255.128 22.99.24.146

#To Divisional Exam Ctrl (Primary)

ip route 22.99.16.0 255.255.254.0 22.99.24.138

#To Divisional Monitoring (Primary)

ip route 22.99.18.0 255.255.254.0 22.99.24.138

#To Divisional (Backup)

ip route 22.99.16.0 255.255.254.0 22.99.24.142 10

ip route 22.99.18.0 255.255.254.0 22.99.24.142 10

#To District Admin via Divisional (Primary)

ip route 22.99.8.0 255.255.252.0 22.99.24.138

#To District Exam via Divisional (Primary)

ip route 22.99.12.0 255.255.252.0 22.99.24.138

#To District via Divisional (Backup)

ip route 22.99.8.0 255.255.252.0 22.99.24.142 10

ip route 22.99.12.0 255.255.252.0 22.99.24.142 10

#To Print Zone via Divisional (Primary)

ip route 22.99.20.0 255.255.254.0 22.99.24.138

#To Packaging via Divisional (Primary)

ip route 22.99.22.0 255.255.254.0 22.99.24.138

#To Printing Press via Divisional (Backup)

ip route 22.99.20.0 255.255.254.0 22.99.24.142 10

ip route 22.99.22.0 255.255.254.0 22.99.24.142 10

#To Backbone via Divisional (Primary)

ip route 22.99.24.128 255.255.255.248 22.99.24.138

#To Backbone via Divisional (Backup)

ip route 22.99.24.128 255.255.255.248 22.99.24.142 10

enable secret class

service password-encryption

end

copy running-config startup-config

DIV-Router

```
enable
configure terminal
hostname DIV-Router
```

```
#Exam Control Cell LAN (22.99.16.0/23)
interface GigabitEthernet0/0
ip address 22.99.16.1 255.255.254.0
no shutdown
```

```
#Monitoring Cell LAN (22.99.18.0/23)
interface GigabitEthernet0/1
ip address 22.99.18.1 255.255.254.0
no shutdown
```

```
#Backbone to Backbone-Switch (22.99.24.128/29)
interface GigabitEthernet0/2
ip address 22.99.24.129 255.255.255.248
no shutdown
```

```
#WAN Primary from NEB (22.99.24.136/30)
interface Serial0/0/0
ip address 22.99.24.138 255.255.255.252
no shutdown
```

```
#WAN Backup from NEB (22.99.24.140/30)
interface Serial0/0/1
ip address 22.99.24.142 255.255.255.252
no shutdown
```

```
#STATIC TO NEB
ip route 22.99.0.0 255.255.248.0 22.99.24.137
ip route 22.99.0.0 255.255.248.0 22.99.24.141 10
```

```
#STATIC TO EDC via NEB
ip route 22.99.24.0 255.255.255.128 22.99.24.137
ip route 22.99.24.0 255.255.255.128 22.99.24.141 10
```

```
#STATIC TO NEB-EDC WAN LINK (Return Path)
ip route 22.99.24.144 255.255.255.252 22.99.24.137
ip route 22.99.24.144 255.255.255.252 22.99.24.141 10
```

#RIPv2 TO DISTRICT & PRINTING PRESS

```
router rip
version 2
no auto-summary
redistribute static
network 22.0.0.0
passive-interface Serial0/0/0
passive-interface Serial0/0/1
passive-interface GigabitEthernet0/0
passive-interface GigabitEthernet0/1
```

#ROUTER-BASED DHCP

```
ip dhcp pool DIV_EXAM_CTRL
network 22.99.16.0 255.255.254.0
default-router 22.99.16.1
dns-server 22.99.0.2
domain-name divisional.gov.bd
```

```
ip dhcp pool DIV_MONITORING
network 22.99.18.0 255.255.254.0
default-router 22.99.18.1
dns-server 22.99.0.2
domain-name divisional.gov.bd
```

```
ip dhcp excluded-address 22.99.16.1 22.99.16.10
ip dhcp excluded-address 22.99.18.1 22.99.18.10
```

```
enable secret class
service password-encryption
end
copy running-config startup-config
```

DIS-Router

```
enable
configure terminal
hostname DIS-Router
```

```
#Admin Unit LAN (22.99.8.0/22)
interface GigabitEthernet0/0
```

```
ip address 22.99.8.1 255.255.252.0
no shutdown
```

```
#Exam Cell LAN (22.99.12.0/22)
interface GigabitEthernet0/1
ip address 22.99.12.1 255.255.252.0
no shutdown
```

```
#Backbone to Backbone-Switch (22.99.24.128/29)
interface GigabitEthernet0/2
ip address 22.99.24.130 255.255.255.248
no shutdown
```

```
#RIPv2
router rip
version 2
no auto-summary
network 22.0.0.0
passive-interface GigabitEthernet0/0
passive-interface GigabitEthernet0/1
```

```
#ROUTER-BASED DHCP
ip dhcp pool DIS_ADMIN
network 22.99.8.0 255.255.252.0
default-router 22.99.8.1
dns-server 22.99.0.2
domain-name district.gov.bd
```

```
ip dhcp pool DIS_EXAM
network 22.99.12.0 255.255.252.0
default-router 22.99.12.1
dns-server 22.99.0.2
domain-name district.gov.bd
```

```
ip dhcp excluded-address 22.99.8.1 22.99.8.10
ip dhcp excluded-address 22.99.12.1 22.99.12.10
```

```
enable secret class
service password-encryption
end
```

```
copy running-config startup-config
```

PRINT-Router

```
enable
```

```
configure terminal
```

```
hostname PRINT-Router
```

```
#Printing Zone LAN (22.99.20.0/23)
```

```
interface GigabitEthernet0/0
```

```
ip address 22.99.20.1 255.255.254.0
```

```
no shutdown
```

```
#Packaging Unit LAN (22.99.22.0/23)
```

```
interface GigabitEthernet0/1
```

```
ip address 22.99.22.1 255.255.254.0
```

```
no shutdown
```

```
#Backbone to Backbone-Switch (22.99.24.128/29)
```

```
interface GigabitEthernet0/2
```

```
ip address 22.99.24.131 255.255.255.248
```

```
no shutdown
```

```
#RIPv2
```

```
router rip
```

```
version 2
```

```
no auto-summary
```

```
network 22.0.0.0
```

```
passive-interface GigabitEthernet0/0
```

```
passive-interface GigabitEthernet0/1
```

```
#STATIC TO NEB (Primary via Divisional)
```

```
ip route 22.99.0.0 255.255.248.0 22.99.24.129
```

```
#FLOATING STATIC (AD = 16 via District)
```

```
ip route 22.99.0.0 255.255.248.0 22.99.24.130 16
```

```
#STATIC TO EDC (for return path)
```

```
ip route 22.99.24.0 255.255.255.128 22.99.24.129
```

```
ip route 22.99.24.144 255.255.255.252 22.99.24.129
```

```
#DHCP RELAY for Packaging Unit
interface GigabitEthernet0/1
ip helper-address 22.99.20.2
```

```
enable secret class
service password-encryption
end
copy running-config startup-config
```

EDC-Router

```
enable
configure terminal
hostname EDC-Router
```

```
#LAN to EDC-Switch (22.99.24.0/25)
interface GigabitEthernet0/0
ip address 22.99.24.1 255.255.255.128
no shutdown
```

```
#WAN to NEB (22.99.24.144/30)
interface Serial0/0/0
ip address 22.99.24.146 255.255.255.252
no shutdown
```

```
#RECURSIVE STATIC ROUTING TO NEB
ip route 22.99.0.0 255.255.248.0 22.99.0.1
ip route 22.99.0.1 255.255.255.255 22.99.24.145
```

```
#Static routes to all other networks via NEB
ip route 22.99.8.0 255.255.252.0 22.99.24.145
ip route 22.99.12.0 255.255.252.0 22.99.24.145
ip route 22.99.16.0 255.255.254.0 22.99.24.145
ip route 22.99.18.0 255.255.254.0 22.99.24.145
ip route 22.99.20.0 255.255.254.0 22.99.24.145
ip route 22.99.22.0 255.255.254.0 22.99.24.145
ip route 22.99.24.128 255.255.255.248 22.99.24.145
```

```
enable secret class
service password-encryption
end
```

copy running-config startup-config

Server Part:

NEB-DNS-Web-Server (Static IP: 22.99.0.2/21, Gateway: 22.99.0.1, DNS: 22.99.0.2)

DNS Service:

1. Turn ON
2. Add Records:
 - a. www.educationboard.gov.bd, A Record, 22.99.0.2
 - b. mail.neb.gov.bd, A Record, 22.99.0.3
 - c. mail.print.gov.bd, A Record, 22.99.20.3
 - d. print.gov.bd, A Record, 22.99.20.3
 - e. neb.gov.bd, A Record, 22.99.0.3

HTTP Service:

1. Turn ON
2. Edit index.html; write: National Examination Control System LIVE

NEB-Mail-Server (Static IP: 22.99.0.3/21)

Email Service:

1. Turn ON
2. Domain: neb.gov.bd
3. SMTP: ON, POP3: ON
4. Add user accounts

PRINT-DHCP-Server — Static IP: 22.99.20.2/23, Gateway: 22.99.20.1, DNS: 22.99.0.2

DHCP Service:

1. Turn ON
2. serverPool (Printing Zone):
 - a. Default Gateway: 22.99.20.1
 - b. DNS Server: 22.99.0.2
 - c. Start IP: 22.99.20.11
 - d. Subnet Mask: 255.255.254.0
 - e. Max Users: 500
3. Pool 1 (Packaging Unit):
 - a. Default Gateway: 22.99.22.1
 - b. DNS Server: 22.99.0.2
 - c. Start IP: 22.99.22.11
 - d. Subnet Mask: 255.255.254.0
 - e. Max Users: 500

PRINT-Mail-Server (Static IP: 22.99.20.3/23, Gateway: 22.99.20.1, DNS: 22.99.0.2)

Email Service:

1. Turn ON
2. Domain: print.gov.bd
3. SMTP: ON, POP3: ON
4. Add user accounts

Validation:

NEB To EDC

```
NEB-Router>ping 22.99.24.146

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 22.99.24.146, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/41/54 ms
```

NEB To DIV Primary

```
NEB-Router>ping 22.99.24.138

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 22.99.24.138, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/38/49 ms
```

NEB To DIV Backup

```
NEB-Router>ping 22.99.24.142

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 22.99.24.142, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/35/54 ms
```

NEB To Server

```
NEB-Router>ping 22.99.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 22.99.0.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

NEB Router to Divisional LAN

```
NEB-Router>ping 22.99.16.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 22.99.16.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 19/35/56 ms
```

NEB Router to District LAN

```
NEB-Router>ping 22.99.8.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 22.99.8.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/30/64 ms
```

NEB Router to Print Zone

```
NEB-Router>ping 22.99.20.1
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 22.99.20.1, timeout is 2 seconds:
```

```
!!!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/32/52 ms
```

NEB Router to EDC LAN

```
NEB-Router>ping 22.99.24.1
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 22.99.24.1, timeout is 2 seconds:
```

```
!!!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/18/42 ms
```

EDC Router to NEB Server

```
EDC-Router>ping 22.99.0.2
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 22.99.0.2, timeout is 2 seconds:
```

```
!!!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/22/42 ms
```

DIS To DNS Server

```
DIS-Router>ping 22.99.0.2
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 22.99.0.2, timeout is 2 seconds:
```

```
!!!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```

DIV Router Route

```

DIV-Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    22.0.0.0/8 is variably subnetted, 17 subnets, 7 masks
S       22.99.0.0/21 [1/0] via 22.99.24.137
R       22.99.8.0/22 [120/1] via 22.99.24.130, 00:00:22, GigabitEthernet0/2
R       22.99.12.0/22 [120/1] via 22.99.24.130, 00:00:22, GigabitEthernet0/2
C       22.99.16.0/23 is directly connected, GigabitEthernet0/0
L       22.99.16.1/32 is directly connected, GigabitEthernet0/0
C       22.99.18.0/23 is directly connected, GigabitEthernet0/1
L       22.99.18.1/32 is directly connected, GigabitEthernet0/1
R       22.99.20.0/23 [120/1] via 22.99.24.131, 00:00:15, GigabitEthernet0/2
R       22.99.22.0/23 [120/1] via 22.99.24.131, 00:00:15, GigabitEthernet0/2
S       22.99.24.0/25 [1/0] via 22.99.24.137
C       22.99.24.128/29 is directly connected, GigabitEthernet0/2
L       22.99.24.129/32 is directly connected, GigabitEthernet0/2
C       22.99.24.136/30 is directly connected, Serial0/0/0
L       22.99.24.138/32 is directly connected, Serial0/0/0
C       22.99.24.140/30 is directly connected, Serial0/0/1
L       22.99.24.142/32 is directly connected, Serial0/0/1
S       22.99.24.144/30 [1/0] via 22.99.24.137

```

PRINT-Router Route

```

PRINT-Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    22.0.0.0/8 is variably subnetted, 15 subnets, 7 masks
S       22.99.0.0/21 [1/0] via 22.99.24.129
R       22.99.8.0/22 [120/1] via 22.99.24.130, 00:00:20, GigabitEthernet0/2
R       22.99.12.0/22 [120/1] via 22.99.24.130, 00:00:20, GigabitEthernet0/2
R       22.99.16.0/23 [120/1] via 22.99.24.129, 00:00:00, GigabitEthernet0/2
R       22.99.18.0/23 [120/1] via 22.99.24.129, 00:00:00, GigabitEthernet0/2
C       22.99.20.0/23 is directly connected, GigabitEthernet0/0
L       22.99.20.1/32 is directly connected, GigabitEthernet0/0
C       22.99.22.0/23 is directly connected, GigabitEthernet0/1
L       22.99.22.1/32 is directly connected, GigabitEthernet0/1
S       22.99.24.0/25 [1/0] via 22.99.24.129
C       22.99.24.128/29 is directly connected, GigabitEthernet0/2
L       22.99.24.131/32 is directly connected, GigabitEthernet0/2
R       22.99.24.136/30 [120/1] via 22.99.24.129, 00:00:00, GigabitEthernet0/2
R       22.99.24.140/30 [120/1] via 22.99.24.129, 00:00:00, GigabitEthernet0/2
S       22.99.24.144/30 [1/0] via 22.99.24.129

```

EDC To NEB Traceroute

```
EDC-Router>tracert 22.99.0.2
Type escape sequence to abort.
Tracing the route to 22.99.0.2

 1  22.99.24.145    1 msec    0 msec    28 msec
 2  22.99.0.2       11 msec    0 msec    0 msec
EDC-Router>|
```

From a PC to NEB DNS Server

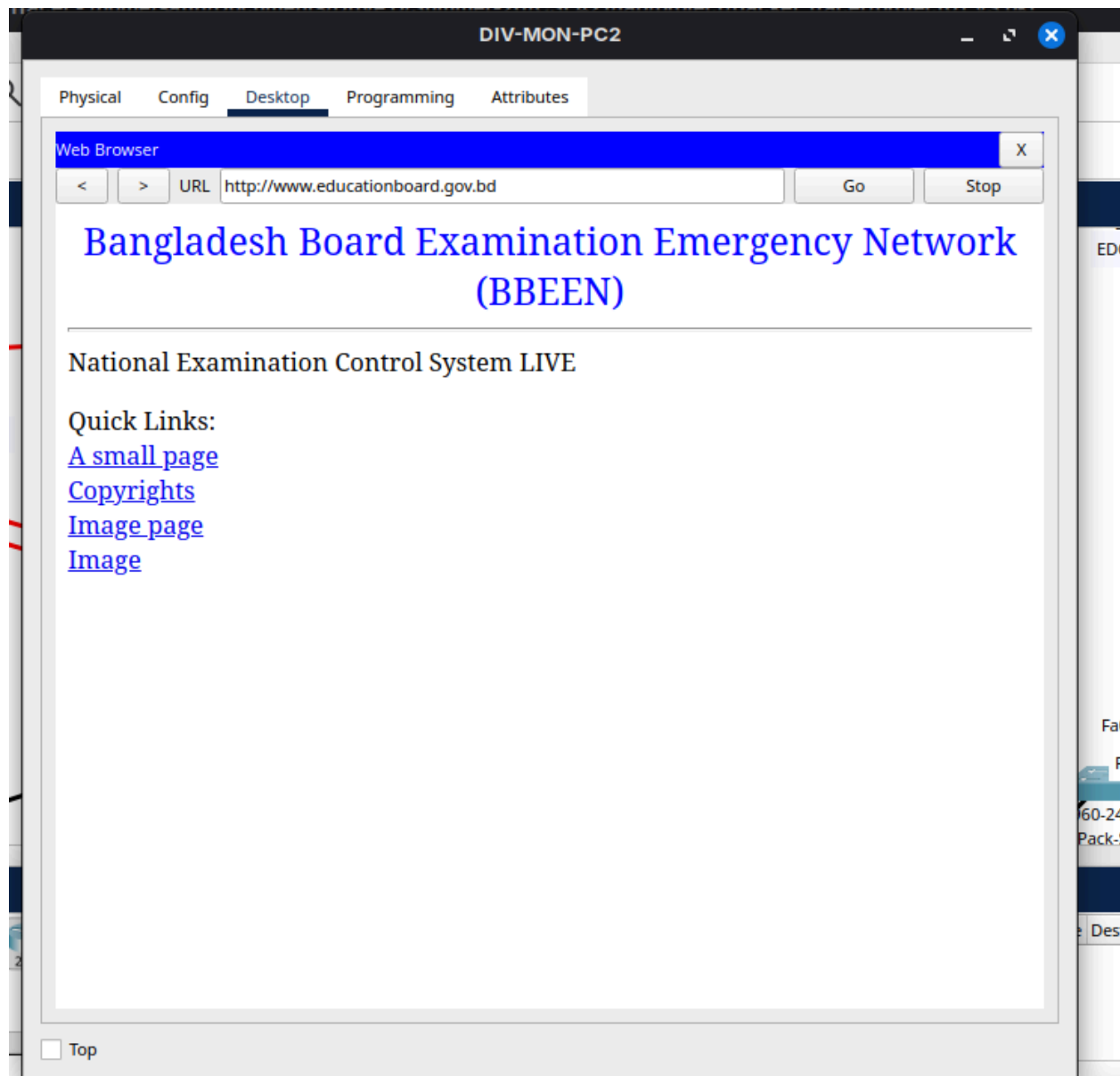
```
C:\>ping 22.99.0.2

Pinging 22.99.0.2 with 32 bytes of data:

Reply from 22.99.0.2: bytes=32 time<1ms TTL=128
Reply from 22.99.0.2: bytes=32 time<1ms TTL=128
Reply from 22.99.0.2: bytes=32 time=9ms TTL=128
Reply from 22.99.0.2: bytes=32 time<1ms TTL=128

Ping statistics for 22.99.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 2ms
```

Divisional PC website access



Email Send and Receive

